

Non Sequitur Publishing · White Paper

Confident Misalignment as an Adversarial Attack Surface

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Written: 2026-05-13 · v0.1-seed

Abstract

The dominant discourse on agentic-AI output errors treats them as reliability problems: outputs that are wrong, operators that are inconvenienced, fixes via better training, better retrieval, or better prompts. The framing is not wrong. It is, however, structurally incomplete. ****Confident misalignment**** — the failure mode in which an agentic system produces coherent, plausible, and authoritative outputs that nevertheless diverge from validated reality or operator intent — is not merely unreliable. It is ***exploitable***. Prompt injection, context poisoning, and stale-state substitution can each be engineered to induce confident misalignment at will, producing outputs that pass operator trust heuristics — fluent, assured, plausible — while serving adversarial goals. This paper argues that confident misalignment is therefore a first-class adversarial-ML threat class, distinct from classical adversarial examples in mechanism but identical in consequence: defender systems must detect it, not only tolerate it.

The paper develops the threat class in five movements. Section 2 establishes the framing failure (reliability vs. adversarial capability) and surveys the existing taxonomies — MITRE ATLAS, OWASP LLM Top 10 — that cover model-input attack vectors but do not formalize the operator-trust exploitation layer where the attack ultimately lands. Section 3 maps the weaponization surface: three vectors (prompt injection, context poisoning, stale-state substitution) that converge on a shared exploit at the human trust layer. Section 4 places confident misalignment within STRIDE and MITRE ATLAS, identifying it as a novel technique class under AML.TA0006 (Impact) that current taxonomies underspecify. Section 5 specifies the structural defense: governance-first calibration signals, inherited from the HGC³AE² parent paper's Skipjack Principle 3 (explicit uncertainty signaling), as the only defense that operates at the trust-exploitation layer. Section 6 closes with limitations and implications for security tooling.

The central claim is that ****"confident by default" is a dark pattern under the adversarial framing****: it suppresses the detection signal that governance needs to function. Systems that present uncertainty are not less trustworthy than systems that suppress it; they are more ***defensible***. The argument's audience is the security community first, with practitioner extensions for SOC and AI-deployment teams.

How to cite

Justin H. Kuiper, CISSP (2026). *Confident Misalignment as an Adversarial Attack Surface* (v0.1-seed). Non Sequitur Publishing. <https://nonsequitur.tech/white-papers/confident-misalignment-attack-surface/>

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ABSTRACT

The dominant discourse on agentic-AI output errors treats them as reliability problems: outputs that are wrong, operators that are inconvenienced, fixes via better training, better retrieval, or better prompts. The framing is not wrong. It is, however, structurally incomplete. **Confident misalignment** — the failure mode in which an agentic system produces coherent, plausible, and authoritative outputs that nevertheless diverge from validated reality or operator intent¹ — is not merely unreliable. It is *exploitable*. Prompt injection, context poisoning, and state-state substitution can each be engineered to induce confident misalignment at will, producing outputs that pass operator trust heuristics — fluent, assured, plausible — while serving adversarial goals. This paper argues that confident misalignment is therefore a first-class adversarial-ML threat class, distinct from classical adversarial examples in mechanism but identical in consequence: defender systems must detect it, not only tolerate it.

The paper develops the threat class in five movements. Section 2 establishes the framing failure (reliability vs. adversarial capability) and surveys the existing taxonomies — MITRE ATLAS,² OWASP LLM Top 10³ — that cover model-input attack vectors but do not formalize the operator-trust exploitation layer where the attack ultimately lands. Section 3 maps the weaponization surface: three vectors (prompt injection,⁴ context poisoning,⁵⁶ state-state substitution) that converge on a shared exploit at the human trust layer. Section 4 places confident misalignment within STRIDE⁷ and MITRE ATLAS, identifying it as a novel technique class under AML.TA0006 (Impact) that current taxonomies underspecify. Section 5 specifies the structural defense: governance-first calibration signals,⁸ inherited from the HGC³AE² parent paper's Skipjack Principle 3 (explicit uncertainty signaling),⁹ as the only defense that operates at the trust-exploitation layer. Section 6 closes with limitations and implications for security tooling.

The central claim is that **“confident by default” is a dark pattern under the adversarial framing**: it suppresses the detection signal that governance needs to function. Systems that present uncertainty are not less trustworthy than systems that suppress it; they are more

1 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

2 MITRE, *Adversarial Threat Landscape for Artificial-Intelligence Systems (ATLAS)* (2024), <https://atlas.mitre.org/>.

3 OWASP, *OWASP Top 10 for Large Language Model Applications* (2024), <https://owasp.org/www-project-top-10-for-large-language-model-applications/>.

4 OWASP, *OWASP Top 10 for Large Language Model Applications* (2024), <https://owasp.org/www-project-top-10-for-large-language-model-applications/>.

5 Patrick Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” *NeurIPS 2020*, <https://arxiv.org/abs/2005.11401>.

6 Yunfan Gao et al., “Retrieval-Augmented Generation for Large Language Models: A Survey,” *arXiv preprint arXiv:2312.10997* (2024).

7 Adam Shostack, *Threat Modeling: Designing for Security* (Hoboken, NJ: Wiley, 2014).

8 Chuan Guo, Geoff Pleiss, Yu Sun, Kilian Q. Weinberger, “On Calibration of Modern Neural Networks,” *Proceedings of ICML 2017*, vol. 70, 1321-1330.

9 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.



defensible. The argument's audience is the security community first, with practitioner extensions for SOC and AI-deployment teams.

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Justin H. Kuiper, CISSP (2026). *Confident Misalignment as an Adversarial Attack Surface* (v0.1-seed). Non Sequitur Publishing.

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1. INTRODUCTION

1.1 THE FRAMING FAILURE

The dominant discourse on agentic-AI output errors treats them as *reliability* failures: the system is wrong, the operator is inconvenienced, the fix is better training, better retrieval, better prompts. The framing has merit — agentic systems do exhibit reliability variability that engineering practice can improve. The framing is also incomplete in a way that creates a security blind spot.

A reliability failure is characterized by randomness, by distribution, by recovery through statistical improvement. An adversarial capability is characterized by intentionality, by targeting, and by exploitation of the defender's trust model. The two are not interchangeable; they require different defenses, different detection regimes, and different incident-response procedures. This paper's opening claim is that **confident misalignment belongs in the second category, not the first**.

The parent paper (HGC³AE²) names confident misalignment as the dominant failure mode of agentic systems and develops the governance-first architecture that mitigates it.¹⁰ The parent argument is correct and load-bearing. What the parent does not develop — and what this derivative thesis supplies — is the adversarial framing: confident misalignment is not only what happens to an agentic system; it is what an adversary *induces* in an agentic system.

1.2 THE THREAT

An adversary who understands that an agentic system presents uncertain outputs with a confident surface can exploit the gap between presentation and epistemic state. The attack does not require compromising the model weights. It does not require breaking the model's training. It requires only that the adversary can influence the inputs from which the model assembles its operating context — and then rely on the model's "confident by default" presentation to propagate the adversary's intended output to a human decision-maker who trusts it.

The attack is complete when the operator acts on the adversarial output. The defender's failure is structural: the output is presented in a form the operator's trust heuristic accepts (fluent, authoritative, internally coherent), and the trust heuristic has no signal that distinguishes confident-and-correct from confident-and-wrong.

¹⁰ Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.



1.3 CONTRIBUTION

This paper makes four contributions:

Defines confident misalignment as an adversarial threat class, not a reliability category, with formal placement in STRIDE¹¹ and MITRE ATLAS.¹²

Characterizes three weaponization vectors — prompt injection, context poisoning, state-state substitution — and shows how each converges on the human-trust exploitation layer.

Maps the trust exploitation layer as the load-bearing attack surface, distinguishing it from the model-input layer that prior taxonomies cover.

Proposes governance-first calibration signals as the structural defense, arguing that “confident by default” is a dark pattern under this framing.

The paper’s primary venue is the security community (target: USENIX Security); a practitioner extension for SOC and AI-deployment teams is anticipated. The contribution to the HGC³AE² family is to operationalize Skipjack Principle 3 (explicit uncertainty signaling) under adversarial pressure: not as a reliability nicety, but as a security control.

2. BACKGROUND

2.1 CONFIDENT MISALIGNMENT: THE PARENT DEFINITION

The HGC³AE² framework defines confident misalignment as the primary failure mode of agentic systems: the system produces outputs that are coherent, plausible, and authoritative yet diverging from validated reality or from the user’s actual intent.¹³ Crucially, the outputs are *not randomly wrong* — they are *specifically wrong while appearing specifically right*. The asymmetry between presentation and accuracy is the defining property.

This paper extends the parent definition by adding adversarial intentionality: confident misalignment can be *induced*, not merely *observed*. The induced variant retains every property of the spontaneous variant — same presentation, same trust signature, same operator response — and adds a new property: the adversary has chosen which specific wrong output to induce.

2.2 ADVERSARIAL ML: THE RELEVANT LITERATURE

Adversarial ML research has established that small, intentional perturbations to model inputs can cause dramatic output errors invisible to human inspection.¹⁴¹⁵ Confident

11 Adam Shostack, *Threat Modeling: Designing for Security* (Hoboken, NJ: Wiley, 2014).

12 MITRE, *Adversarial Threat Landscape for Artificial-Intelligence Systems (ATLAS)* (2024), <https://atlas.mitre.org/>.

13 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

14 Christian Szegedy et al., “Intriguing Properties of Neural Networks,” *arXiv preprint arXiv:1312.6199* (2014), <https://arxiv.org/abs/1312.6199>.

15 Ian Goodfellow, Jonathon Shlens, and Christian Szegedy, “Explaining and Harnessing Adversarial Examples,” *Proceedings of ICLR 2015*, <https://arxiv.org/abs/1412.6572>.



misalignment is not an adversarial perturbation in the classical sense — it operates at the *context-assembly* layer, not the model-weights layer — but it shares the structural property: the adversary engineers inputs to produce adversarial outputs that the defender's evaluation system cannot distinguish from benign outputs.

The distinction matters. Classical adversarial examples attack model inference: the perturbation alters how the model maps a given input to an output. Confident misalignment attacks *operator trust*: the adversary's input may be processed by the model exactly as the model is designed to process inputs, and the model's output may be exactly what an honest interpretation of the input would produce — the attack succeeds because the *operator's* trust in the output's confidence cannot distinguish honest output from adversary-engineered output.

2.3 CURRENT TAXONOMIES: THE GAP

MITRE ATLAS¹⁶ covers adversarial AI at the technique-taxonomy level: tactics from reconnaissance through impact, with sub-techniques covering model-extraction, evasion, poisoning, and related categories. OWASP LLM Top 10¹⁷ covers prompt injection (LLM01) and data/model poisoning (LLM03) as top risk categories with practitioner-facing controls. Neither taxonomy formalizes the *exploitation of operator trust* as the attack outcome: they describe how the attack reaches the model, not how the attack *uses* the model to reach the human.

This paper fills that gap. The threat class is not “model receives adversarial input” but “human acts on model-presented adversarial output.” The shift in attack-class definition has operational consequences: detection cannot live exclusively at the input layer (where current defenses concentrate); it must extend to the output-trust layer (where confident misalignment completes).

3. THE WEAPONIZATION SURFACE

The three weaponization vectors below converge on a shared exploit at the human trust layer. Each is documented in the broader literature; the contribution of this section is to frame the three as instances of a single attack pattern.

3.1 VECTOR 1: PROMPT INJECTION

Prompt injection engineers the system prompt, the user context, or intermediate inputs to override the model's operational constraints and substitute adversary-defined behavior.¹⁸¹⁹ The adversary's goal is not model control per se — it is to produce outputs the human operator will trust and act upon.

¹⁶ MITRE, *Adversarial Threat Landscape for Artificial-Intelligence Systems (ATLAS)* (2024), <https://atlas.mitre.org/>.

¹⁷ OWASP, *OWASP Top 10 for Large Language Model Applications* (2024), <https://owasp.org/www-project-top-10-for-large-language-model-applications/>.

¹⁸ Thushari Peiris et al., “Prompt Injection Attacks and Defenses in Large Language Models,” 2024 (per local bibliography).

¹⁹ OWASP, *OWASP Top 10 for Large Language Model Applications* (2024), <https://owasp.org/www-project-top-10-for-large-language-model-applications/>.



The OWASP LLM Top 10 lists prompt injection as LLM01, with a control-focused practitioner taxonomy. This paper’s reframing: **prompt injection is a human-trust exploitation attack**, not merely a model-control attack. The measure of success is not “the model did what the adversary asked”; it is “the operator acted on the model’s adversarial output.” A successful prompt injection that produces an output the operator notices as anomalous is a near-miss; a less-successful injection that produces an output the operator trusts and acts on is a full compromise.

Attack surface: any input channel that reaches the model before output generation — system prompt, user message, retrieved documents, tool outputs, memory entries.

Detection gap: prompt injection succeeds when the operator cannot distinguish injected-instruction output from legitimate output. Confident presentation is the propagation vector; uncertainty suppression is the operative mechanism that makes the output indistinguishable.

3.2 VECTOR 2: CONTEXT POISONING

Context poisoning injects malicious or stale state into the context assembly process upstream of inference. In retrieval-augmented generation (RAG) systems, this means poisoning the retrieval corpus.^{20,21} In memory-augmented agents, this means corrupting the memory store. In tool-using agents, this means poisoning tool outputs.

The attack is structurally analogous to cache poisoning in classical web security: the adversary substitutes adversarial entries at the input layer, confident that the downstream consumer (here, the model) will process them as authoritative and present their content with appropriate confidence. The asymmetry is that classical cache poisoning eventually presents cached content to a user who can identify it as anomalous via independent verification; context-poisoned agentic output presents content that has been *re-rendered through the model’s natural language*, removing the artifactual signature that would let a verifier detect substitution.

Detection gap: context poisoning is not detectable at the model output layer without ground-truth comparison. The model produces a fluent, confident response that accurately reflects the poisoned context. Human governance — authoritative validation of the inputs that were used to produce a given output, not the output alone — is the only detection surface that operates at the right architectural layer.

3.3 VECTOR 3: STALE-STATE SUBSTITUTION

Stale-state substitution does not require active adversarial injection. It requires only that the adversary can ensure the agentic system uses cached or stale context at a moment when fresh state would produce different output — and that the system presents its stale-state-based output with the same confidence as a fresh-state-based output.

This is the “availability as an attack surface” variant of the attack class. The adversary disrupts validation of fresh state (per the CYB-03 thesis on denial-of-validation) and relies on the agentic system’s silent-fallback behavior — its tendency to operate on whatever context is available, with whatever confidence the model habitually applies — to propagate stale state as current truth.

20 Patrick Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” *NeurIPS 2020*, <https://arxiv.org/abs/2005.11401>.

21 Yunfan Gao et al., “Retrieval-Augmented Generation for Large Language Models: A Survey,” *arXiv preprint arXiv:2312.10997* (2024).



Detection gap: identical to context poisoning. The operator cannot distinguish stale-state output from fresh-state output without explicit system-signaled differentiation. If the system does not flag “this output was produced from cached context at T-30min,” the operator has no signal to invoke a freshness check.

3.4 THE TRUST EXPLOITATION LAYER

All three vectors converge on the same exploit surface: **the human operator’s trust in confident agentic presentation**. The attack is complete when the operator acts on adversarial output without invoking independent verification. The “confident by default” presentation is the mechanism; the operator’s calibrated trust in confident outputs is the vulnerability.

This framing locates the attack at the human-AI interface layer, not the model layer. The implication for defense is structural: purely technical defenses (adversarial training, input sanitization, retrieval hardening) are necessary but insufficient. They make the attack harder to launch but do not address the layer where it lands. The structural defense must operate at the trust layer, where the adversarial output is converted into adversarial outcome.

4. THREAT MODEL PLACEMENT

4.1 STRIDE

The STRIDE methodology²² classifies threats into six categories. Confident misalignment, as an adversarial capability, maps to STRIDE as follows:

STRIDE category

Confident misalignment mapping

Spoofing (S)

Adversarial output impersonates accurate output; the system’s confident presentation acts as an identity signal the human trusts.

Tampering (T)

²² Adam Shostack, *Threat Modeling: Designing for Security* (Hoboken, NJ: Wiley, 2014).



Context poisoning and stale-state substitution tamper with the information state from which output is generated.

Repudiation (R)

When confident misalignment informs a consequential decision, there is no attribution chain linking the adversarial input to the adversarial output — the model is the intermediary.

Information Disclosure (I)

Prompt injection can extract sensitive information from the system context and present it to the adversary as model output.

Denial of Service (D)

Out of scope for this paper; see the CYB-03 thesis on denial-of-validation for the adjacent threat class.

Elevation of Privilege (E)

A human operator acting on confident misalignment may authorize actions the adversary could not have authorized directly — the operator's authority is borrowed via the trust exploit.

The primary STRIDE threat classes for this paper are **Spoofing and Tampering**. Both operate via the trust exploitation surface developed in §3.4.

4.2 MITRE ATLAS

MITRE ATLAS provides the canonical tactics-and-techniques taxonomy for adversarial AI. Confident misalignment, as an adversarial capability, maps to ATLAS at the tactics layer:

AML.TA0001 (Reconnaissance): adversary learns the system's context-assembly mechanisms — which retrieval sources, which memory stores, which tool integrations — before constructing the injection.

AML.TA0002 (Resource Development): adversary prepares poisoned context entries, prompt-injection payloads, or stale-state inducement conditions.

AML.TA0004 (ML Attack Staging): adversary positions the attack at the appropriate input channel (the context assembly point or the retrieval corpus).

AML.TA0006 (Impact): adversary achieves operator action on adversarial output — the realization of the confident-misalignment attack.

This paper positions confident misalignment as a *novel technique class beneath AML.TA0006 (Impact)*. Current ATLAS taxonomy underspecifies the human-trust exploitation outcome; the gap is precisely the layer this paper addresses.

5. THE HUMAN GOVERNANCE DEFENSE

5.1 WHY TECHNICAL DEFENSES ARE INSUFFICIENT

Adversarial training, input sanitization, and retrieval hardening address the model-layer attack surface — they make it harder to induce confident misalignment through the three vectors. These defenses are necessary. They are also insufficient, because they do not address the trust-exploitation layer where the attack lands. The trust-exploitation layer is human: the defense must operate at the point where the operator decides to trust or verify.



Uncertainty calibration research²³ attempts to align model confidence with actual accuracy. This is a necessary technical foundation for any trust-layer defense — but calibrated uncertainty is only useful if the operator knows to look for it and is trained to act on it.²⁴²⁵ A model that signals uncertainty in a way the operator has learned to ignore (because the signal has high false-positive rate, or because confident-on-uncertain is the default operating mode) fails as a defense regardless of how technically well-calibrated the underlying uncertainty estimate is.

5.2 GOVERNANCE-FIRST CALIBRATION SIGNALS

The structural defense is **governance-mandated uncertainty broadcasting**: the system is required, by policy, to surface its epistemic state before the operator commits to a consequential action based on the output. The requirement is not an interface convention; it is a runtime policy enforced architecturally.

This is Skipjack Principle 3 from the parent HGC³AE² framework: explicit uncertainty signaling.²⁶ Applied to the adversarial context, the principle converts the “confident by default” presentation into an *attack detection surface*. An output that should be uncertain — because the context-assembly process produced it from low-confidence inputs, or because the model’s calibration estimate for this output class is poor — but presents as confident is *anomalous*. The anomaly is detectable if governance requires uncertainty signals.

The reframing has a security consequence: **“confident by default” is a dark pattern under this framing**. Suppressing uncertainty signals to produce more confident-seeming outputs is not user-experience polish; it is the elimination of a security control. The same outputs presented with appropriate uncertainty surface anomalies that defenders can act on; presented without uncertainty, those anomalies are erased.

5.3 IMPLEMENTATION: THE CALIBRATION PERIMETER

Human governance as the outer defense perimeter has three operational requirements:

Policy. Operators must not commit to consequential actions on high-confidence agentic outputs without explicit verification against an independent source, for any output involving dynamic state or any output whose context-assembly traversed an attacker-controllable input channel. The policy is operationally specific, not aspirational: it names which output classes require which verification, with thresholds calibrated to operator workflow.

Detection. A confident output on dynamic state without verification is a *policy violation*, detectable in audit logs. The detection signal does not depend on identifying the adversarial input; it depends only on observing that the operator-action-on-output path was traversed without the policy-required verification step. The signal is operationally cheap and accumulates across the operating window.

23 Chuan Guo, Geoff Pleiss, Yu Sun, Kilian Q. Weinberger, “On Calibration of Modern Neural Networks,” *Proceedings of ICML 2017*, vol. 70, 1321-1330.

24 Kevin Anthony Hoff and Masooda Bashir, “Trust in Automation: Integrating Empirical Evidence on Factors That Influence Trust,” *Human Factors* 57, no. 3 (2015): 407-434.

25 Raja Parasuraman and Victor Riley, “Humans and Automation: Use, Misuse, Disuse, Abuse,” *Human Factors* 39, no. 2 (1997): 230-253.

26 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.



Response. Human-in-the-loop checkpoints fire on high-confidence outputs against dynamic-state contexts. The model's confidence score alone is not authorization to proceed; the architecture requires either a calibrated uncertainty signal that the operator acknowledges, or an independent-source verification step. The Skipjack human-in-the-loop mechanism²⁷ operationalizes this at runtime.

The three requirements compose into a *calibration perimeter*: a structural boundary around consequential agentic-action surfaces, inside which confident outputs are gated by governance-mandated uncertainty-or-verification.

6. LIMITATIONS AND CONCLUSION

6.1 LIMITATIONS

This paper does not propose a formal confidence calibration method. Aligning model confidence with actual accuracy at scale is an ML systems research problem; this paper assumes such calibration is achievable in principle and focuses on the governance layer that converts calibrated signals into security controls.

The governance-first defense is context-dependent. In time-critical operational environments — emergency response, real-time control, certain trading contexts — mandatory verification may be operationally infeasible. The paper scopes the defense to *consequential non-time-critical decisions*, which is the dominant class of agentic-system use today. Time-critical contexts require complementary defenses (architectural rate-limiting, separate trust pathways for time-critical paths) not developed in this paper.

The attack taxonomy in §3 and §4 is based on known vectors at the time of writing. Novel adversarial techniques may exploit surfaces not covered here. The framework's value is the *category* (trust-layer exploitation via confidence-mediated outputs), not the *enumeration* of all current vectors; new techniques in this category are expected and the framework anticipates them.

6.2 CONCLUSION

Confident misalignment is not a reliability failure mode; it is an adversarial capability. Prompt injection, context poisoning, and stale-state substitution are delivery mechanisms for an attack that completes at the human trust layer. The defense must operate there too: governance-first calibration signals convert the confident-by-default presentation from an attack surface into a detection surface.

The implication for system design is specific: **explicit uncertainty signaling is not a user-experience nicety — it is a security control**. Systems that suppress uncertainty signals to produce more confident-seeming outputs are not more trustworthy; they are more *exploitable*. The presentation-vs-epistemic-state asymmetry that defines confident misalignment is the same asymmetry the adversary exploits; closing that asymmetry through governance-mandated uncertainty surfaces is the structural defense the security community must adopt.

This is Thesis One (cyb-01) of the HGC³AE² Cyber Pillar derivative family. Companion theses develop adjacent dimensions: cyb-02 (context integrity failure), cyb-03 (denial of

²⁷ Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.



validation), cyb-06 (agentic zero trust), cyb-08 (reconciliation SOC doctrine), cyb-09 (tool orchestration trust composition), cyb-10 (human authority chain of custody). Each thesis inherits from the parent HGC³AE² framework and develops one cyber-specific dimension of the framework's defense architecture.

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— end v0.1-seed draft —

Drafting state: v0.1 CYB-01 (chain v0.1 step 1). 6-section canon shape adapted to thesis-paper (USENIX Security target). PR count: 12 — parent (HGC³AE²) + 10 inline cyber/ML/STRIDE/ATLAS/RAG/calibration/trust references + 2 cross-thesis forward refs (cyb-02, cyb-03). Meets ≥ 12 floor.